SMART INDIA HACKATHON 202

TITLE PAGE

- **Problem Statement ID – SIH26167**
- **Problem Statement Title- SatQuery AI – An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries**
- **Theme- Space Technology**
- **PS Category- Software**
- **Team ID- [fill after portal registration]**
- **Team Name (Registered on portal)- [fill after portal registration]**

SatQuery AI

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IDEA TITLE

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- Problem: analysts manually interpret ISRO imagery today for agri-insurance, land records, and mining compliance – today's workflow is Search → Select → Analyse → Interpret: slow, expertise-heavy, no natural-language access
- Solution: ask in plain language, get an answer grounded in real ISRO evidence – fills the gap between structured dashboards (SatSure) and ungrounded global search (Google Earth AI)

Plain-Language Query → **Agent Routing** → **Retrieve + Analyze (ISRO Archive)** → **Validated, Grounded Answer**

Platform capabilities: semantic retrieval • VQA & visual grounding • confidence scoring — change detection & optical+SAR fusion (roadmap, after MVP validation)

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TECHNICAL APPROACH

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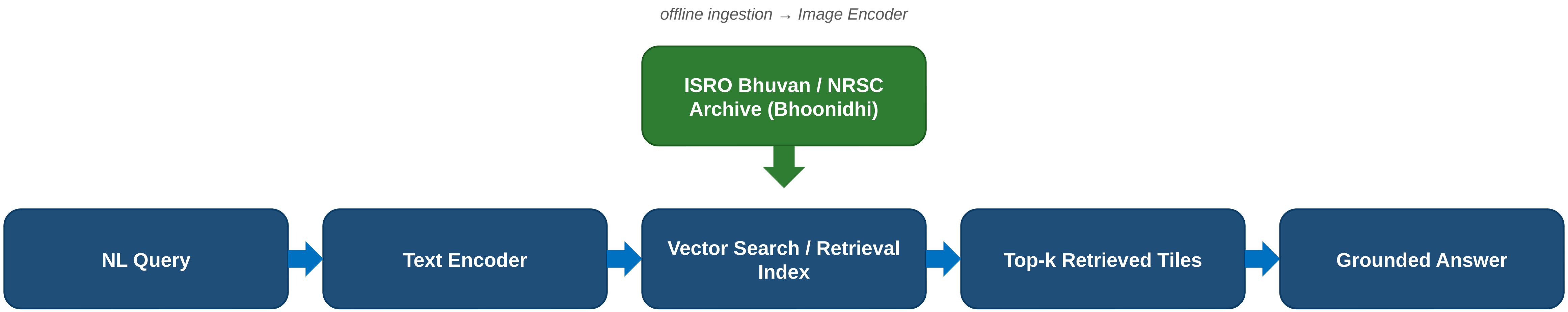
- Stack: Python, PyTorch, RemoteCLIP-family dual encoder (image + text), fine-tuned on BigEarthNet-MM, evaluated on VRSBench
- Pipeline: Query → Text Encoder → Vector Search → Top-k Retrieval → Grounded Answer (flow below), sourced from the ISRO archive



MVP path shown in full; the same agent layer routes to change-detection and optical+SAR specialist models post-validation

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FEASIBILITY AND VIABILITY

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FEASIBILITY

Pretrained encoders + open datasets (BigEarthNet, VRSBench) → fine-tuning, not new invention. Main risk: retrieval accuracy on Indian-specific terrain classes.

RISK

Indian Domain Gap: training data may not fully represent Indian terrain

Multimodal Complexity: optical, SAR & temporal data need different analysis

AI Reliability: plausible answers can still be incorrect

MITIGATION

MVP proves retrieval + grounded VQA end-to-end on one region first – change detection & SAR fusion phased in after validation

Trust: every answer ships with evidence, a confidence score, and an execution trace

BUSINESS MODEL (UNIT ECONOMICS)

Usage-based per-query/report API to agri-insurance, land-record & mining-compliance verticals – priced below enterprise GEOINT contracts (SatSure, Google Cloud), since ISRO archive access is the cost base.

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IMPACT AND BENEFITS

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What this solves: turns a static satellite archive into a decision-ready answer, in plain language

AGRI-INSURANCE

“Does this field show crop damage?”

Impact: Faster claim assessment

LAND RECORDS

“Show construction beyond this boundary.”

Impact: Accessible geospatial analysis

MINING & ENVIRONMENT

“What changed around this mining lease?”

Impact: Faster compliance screening

DISASTER RESPONSE

“Which areas changed after the flood?”

Impact: Faster satellite-to-decision

Bottom line: AI doesn’t replace experts — it accelerates their analysis.

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RESEARCH AND REFERENCES

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- BigEarthNet-MM dataset (arXiv:2603.29630) – primary fine-tuning corpus (Sentinel-1 SAR + Sentinel-2 multispectral, co-registered)
- VRSBench – public evaluation benchmark for remote-sensing vision-language tasks
- Prior art (open research, acknowledged directly): RemoteCLIP, GeoChat, TEOChat – remote-sensing vision-language architectures this design builds on
- Adjacent commercial/global tools (differentiation, not duplication): SatSure SatScore (structured analytics, not conversational); Google Maps Platform Aerial & Satellite Insights (global zero-shot search, no ISRO archive integration)
- Data source: ISRO Bhuvan / NRSC data portal, accessed via Bhoonidhi (Department of Space)